

An Informal Introduction to Function Limits

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Finite limit as x approaches a finite value

Intuitive Definition

The limit of a function $f(x)$ at a point $x = x_0$ describes the value that the function approaches as x gets closer to x_0 .

If $f(x)$ gets closer and closer to a value L as x approaches x_0 **from both sides**, then we say that the limit of $f(x)$ as x approaches x_0 is L , and we write:

$$\lim_{x \rightarrow x_0} f(x) = L$$

Finite limit as x approaches a finite value

Example

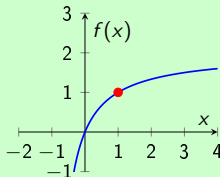
Consider the function $f(x) = \frac{2x}{x+1}$.

$$\lim_{x \rightarrow 1} f(x) = \frac{2 \cdot 1}{1 + 1} = \frac{2}{2} = 1$$

Here, the function approaches the value 1 as x gets close to 1.

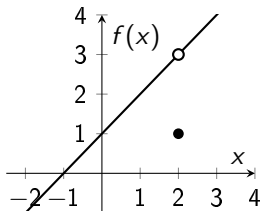
Graphical Interpretation

Looking at the graph of $f(x) = \frac{2x}{x+1}$, we can visually see that the function gets closer to the value 1 as x approaches 1.



Limit and Continuity at a Point

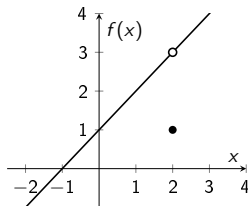
Let $f(x) = x + 1$ for $x \neq 2$, and let $f(2) = 1$. The graph has a hole at $(2, 3)$ and includes the point $(2, 1)$. The function is discontinuous at $x = 2$.



Limit Evaluation

- $\lim_{x \rightarrow -1} f(x) = ?$
- $\lim_{x \rightarrow 0} f(x) = ?$
- $\lim_{x \rightarrow 2} f(x) = ?$

Limit and Continuity at a Point



Limit Results

- $\lim_{x \rightarrow -1} f(x) = 0$
- $\lim_{x \rightarrow 0} f(x) = 1$
- $\lim_{x \rightarrow 2} f(x) = 3$

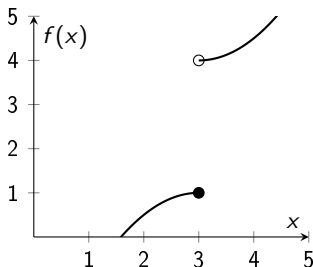
Important Note

We are not asking for the value of the function at $x = 2$, but for the value the function **approaches** as x gets infinitely close to 2.

Left-hand and Right-hand Limits

Let's consider the function $f(x)$, which is defined piecewise:

$$f(x) = \begin{cases} -\frac{1}{2}(x-3)^2 + 1 & \text{if } x \leq 3, \\ \frac{1}{2}(x-3)^2 + 4 & \text{if } x > 3. \end{cases}$$



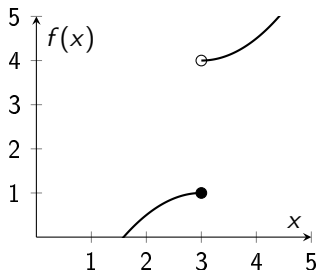
Limit Evaluation

- $\lim_{x \rightarrow 3^-} f(x) = ?$
- $\lim_{x \rightarrow 3^+} f(x) = ?$
- $\lim_{x \rightarrow 3} f(x) = ?$

Left-hand and Right-hand Limits

Limit Evaluation

- $\lim_{x \rightarrow 3^-} f(x) = 1$
- $\lim_{x \rightarrow 3^+} f(x) = 4$

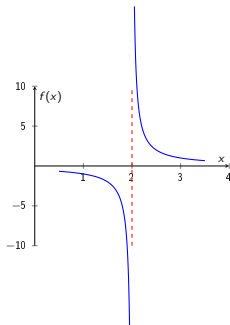


The Limit Does Not Exist

The limit $\lim_{x \rightarrow 3} f(x)$ does not exist, because by definition **a limit exists at a point only if the left-hand and right-hand limits are equal**. This is another example of a discontinuity.

Vertical Asymptotes

Let's consider the function $f(x) = \frac{1}{x-2}$, which has a vertical asymptote at $x = 2$.



Limit Evaluation

- $\lim_{x \rightarrow 2^-} f(x) = -\infty$
- $\lim_{x \rightarrow 2^+} f(x) = +\infty$
- Therefore, $\lim_{x \rightarrow 2} f(x)$ does not exist.

Intro to Limit Arithmetic: Infinite Limits

What happens when a denominator grows without bound?

Imagine dividing a constant, such as 1 or 5, by increasingly large numbers, like 1000, 1000000, or 1000000000. As the denominator increases, the result gets smaller:

$$\frac{1}{1000} = 0.001, \quad \frac{1}{1000000} = 0.000001, \quad \text{and so on.}$$

Eventually, as the denominator tends to infinity, the result gets arbitrarily close to 0. For every fixed real constant k , this is expressed by $\lim_{x \rightarrow +\infty} k/x = 0$. This is a limit, not division by a real number called infinity.

What happens as a positive denominator approaches zero?

Now imagine dividing a constant, like 1, by smaller and smaller numbers: 0.1, 0.01, 0.001, etc. As the denominator gets closer to zero, the result grows rapidly:

$$\frac{1}{0.1} = 10, \quad \frac{1}{0.01} = 100, \quad \frac{1}{0.001} = 1000, \quad \text{and so on.}$$

As the denominator becomes extremely small, the result becomes extremely large, tending to $+\infty$ (if the numerator is positive) or $-\infty$ (if negative). For $k > 0$, $\lim_{x \rightarrow 0^+} k/x = +\infty$ and $\lim_{x \rightarrow 0^-} k/x = -\infty$. Division by zero is undefined.

Infinity in Limit Notation

Unbounded Behavior

The symbols $+\infty$ and $-\infty$ are not real numbers. In a limit, they describe values that eventually exceed every real bound, or fall below every real bound.

Different Mathematical Contexts

Arithmetic with extended real values has specific rules and undefined forms. Cardinal and ordinal arithmetic use different notions of infinity. For example, adding one need not have the same meaning in those two settings.

Division by Zero

A limit involving a denominator approaching zero must be evaluated using its numerator, sign and rate of approach. It does not define division by zero.

Vertical and Horizontal Asymptotes

An asymptote describes limiting behavior of a graph. Here we consider vertical and horizontal asymptotes; oblique asymptotes are another possibility. A graph can cross a horizontal or oblique asymptote.

Vertical Asymptotes

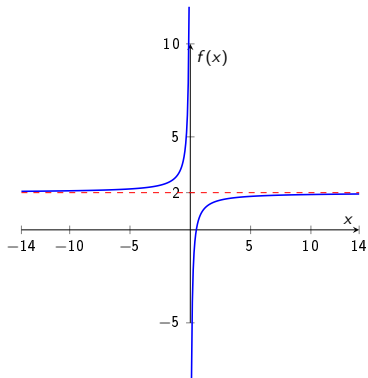
Vertical asymptotes occur when a function tends toward $+\infty$ or $-\infty$ as it approaches a specific point $x = c$. We saw this with the function $f(x) = \frac{1}{x-2}$, which has a vertical asymptote at $x = 2$.

Horizontal Asymptotes

Horizontal asymptotes describe how a function behaves as x approaches $+\infty$ or $-\infty$. A horizontal asymptote exists when $\lim_{x \rightarrow \pm\infty} f(x)$ equals a finite value.

Horizontal and Vertical Asymptotes

Consider the function $f(x) = 2 - \frac{1}{x}$.



$$\lim_{x \rightarrow -\infty} f(x) = ? \quad \text{and} \quad \lim_{x \rightarrow +\infty} f(x) = ?$$

$$\lim_{x \rightarrow 0^+} f(x) = ? \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = ?$$